

Navigation in a Grid-World with Obstacles: Derivation of the Fokker-Planck Equation

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1 Setup and Policies

1.1 State space and actions

We consider an agent navigating a 2D grid-world. The state of the agent is its grid position $\bar{s} = (x, y)$, and the set of available actions is $\mathcal{A} = \{N, S, E, W\}$, corresponding to unit steps of size ϵ along the cardinal directions:

$$+\hat{x}, \quad -\hat{x}, \quad +\hat{y}, \quad -\hat{y}. \quad (1)$$

The goal is located at $\bar{g}$. Some grid cells are occupied by obstacles, and whether the four neighbouring cells are occupied is encoded by the local occupancy vector $\mathcal{N}_s = \{O_N, O_S, O_E, O_W\}$, where $O_i \in \{0, 1\}$ with 0 denoting free and 1 denoting occupied.

1.2 Geocentric policy $\pi_G(a \mid s)$

The geocentric policy biases the agent toward the goal $\bar{g}$. For each action $\bar{a} = \epsilon \hat{e}_a$, the alignment score is the cosine of the angle between the action direction and the goal direction:

$$\phi_a = \cos \theta_{ag} = \frac{\bar{a} \cdot (\bar{g} - \bar{s})}{|\bar{a}| |\bar{g} - \bar{s}|} = \hat{e}_a \cdot \hat{u}, \quad \hat{u} = \frac{\bar{g} - \bar{s}}{|\bar{g} - \bar{s}|}. \quad (2)$$

Writing $\hat{u} = (u_x, u_y)$, the four scores are:

$$\phi_{+x} = u_x, \quad \phi_{-x} = -u_x, \quad \phi_{+y} = u_y, \quad \phi_{-y} = -u_y. \quad (3)$$

The policy is a softmax over these scores with inverse temperature $\beta > 0$:

$$\pi_G(a \mid s) = \frac{\exp[\beta \phi_a]}{\sum_{a'} \exp[\beta \phi_{a'}]}. \quad (4)$$

A large β makes the policy nearly deterministic toward the goal, while $\beta \rightarrow 0$ gives a uniform random walk.

Define the axis-wise partition functions:

$$Z_x^G = e^{\beta u_x} + e^{-\beta u_x} = 2 \cosh(\beta u_x), \quad Z_y^G = e^{\beta u_y} + e^{-\beta u_y} = 2 \cosh(\beta u_y), \quad (5)$$

so that $Z_G = Z_x^G + Z_y^G = 2 \cosh(\beta u_x) + 2 \cosh(\beta u_y)$.

1.3 Egocentric policy $\pi_E(a \mid s)$

The egocentric policy reacts to the local occupancy of the four neighbouring cells. Let $O_a \in \{0, 1\}$ be the occupancy of the cell reached by action a (0 = free, 1 = blocked). The policy penalises blocked directions with a repulsion strength $\alpha > 0$:

$$\pi_E(a \mid s) = \frac{\exp[-\alpha O_a]}{\sum_{a'} \exp[-\alpha O_{a'}]}. \quad (6)$$

A free cell contributes $e^0 = 1$ to the partition function; a blocked cell contributes $e^{-\alpha}$. Defining the axis-wise partition functions:

$$Z_{E_x} = e^{-\alpha O_{+x}} + e^{-\alpha O_{-x}}, \quad Z_{E_y} = e^{-\alpha O_{+y}} + e^{-\alpha O_{-y}}, \quad (7)$$

we have $Z_E = Z_{E_x} + Z_{E_y}$.

If there are $k \in \{0, 1, 2, 3, 4\}$ blocked neighbours in total, then $Z_E = (4 - k) + k e^{-\alpha}$.

1.4 Combined policy

The agent's stochastic policy is a convex mixture of the geocentric and egocentric policies, controlled by the mixing parameter $\gamma \in [0, 1]$:

$$\pi(a \mid s) = \gamma \pi_G(a \mid s) + (1 - \gamma) \pi_E(a \mid s). \quad (8)$$

At $\gamma = 1$ the agent is purely goal-directed; at $\gamma = 0$ it reacts only to local obstacles.

2 From Master Equation to Fokker-Planck

2.1 Master equation

Let $P(\bar{r}, t)$ be the probability of finding the agent at position $\bar{r}$ at time t . The discrete-time evolution is given by the master equation:

$$\frac{\partial P(\bar{r}, t)}{\partial t} = \sum_{\bar{a}} \left[\pi(\bar{a} \mid \bar{r} - \bar{a}) P(\bar{r} - \bar{a}, t) - \pi(\bar{a} \mid \bar{r}) P(\bar{r}, t) \right]. \quad (9)$$

The first term accounts for probability flowing *into* $\bar{r}$ from neighbouring sites; the second accounts for probability flowing *out* of $\bar{r}$.

2.2 Kramers–Moyal expansion

We pass to the continuum limit by Taylor-expanding $\pi(\bar{a} \mid \bar{r} - \bar{a}) P(\bar{r} - \bar{a}, t)$ around $\bar{r}$ in powers of the step size ϵ :

$$\pi(\bar{a} \mid \bar{r} - \bar{a}) P(\bar{r} - \bar{a}, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (\bar{a} \cdot \nabla)^n [\pi(\bar{a} \mid \bar{r}) P(\bar{r}, t)]. \quad (10)$$

Substituting into (9), the $n = 0$ terms cancel (gain equals loss), and we obtain the Kramers–Moyal expansion:

$$\frac{\partial P}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \sum_{i_1, \dots, i_n} \partial_{i_1} \cdots \partial_{i_n} [M_{i_1 \dots i_n}^{(n)}(\bar{r}) P(\bar{r}, t)], \quad (11)$$

where the n -th Kramers–Moyal moment of the policy is:

$$\bar{M}^{(n)}(\bar{r}) = \sum_{\bar{a}} \bar{a}^{\otimes n} \pi(\bar{a} \mid \bar{r}). \quad (12)$$

2.3 Fokker-Planck equation (second-order truncation)

Retaining terms up to $\mathcal{O}(\epsilon^2)$ (i.e. $n = 1$ and $n = 2$) gives the Fokker-Planck equation:

$$\boxed{\frac{\partial P}{\partial t} = -\nabla \cdot [\bar{M}^{(1)}(\bar{r}) P] + \frac{1}{2} \nabla \nabla : [\bar{M}^{(2)}(\bar{r}) P].} \quad (13)$$

Since the moment tensor is linear in π , and $\pi = \gamma \pi_G + (1 - \gamma) \pi_E$, we have:

$$\bar{M}^{(n)} = \gamma \bar{M}_G^{(n)} + (1 - \gamma) \bar{M}_E^{(n)}. \quad (14)$$

We now compute $\bar{M}^{(1)}$ and $\bar{M}^{(2)}$ for each policy separately.

3 Geocentric Moments

3.1 First moment $\bar{M}_G^{(1)}$ (drift)

The first moment is the probability-weighted expected displacement:

$$\bar{M}_G^{(1)} = \sum_{\bar{a}} \bar{a} \pi_G(\bar{a}). \quad (15)$$

Each action contributes a displacement $\pm \epsilon$ along one axis. The $\pm x$ actions have scores $+\beta u_x$ and $-\beta u_x$ respectively, and similarly for y . Since actions along x contribute zero to the y displacement and vice versa, the two axes decouple:

$$\bar{M}_{G,x}^{(1)} = \epsilon [\pi_G(+\hat{x}) - \pi_G(-\hat{x})] = \epsilon \frac{e^{\beta u_x} - e^{-\beta u_x}}{Z_G} = \epsilon \frac{2 \sinh(\beta u_x)}{2 \cosh(\beta u_x) + 2 \cosh(\beta u_y)}. \quad (16)$$

The $\sinh$ arises naturally because the $+\hat{x}$ and $-\hat{x}$ scores are *exactly opposite* by construction of the dot product, so their difference is antisymmetric. This gives:

$$\boxed{\bar{M}_{G,x}^{(1)} = \epsilon \frac{\sinh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)}, \quad \bar{M}_{G,y}^{(1)} = \epsilon \frac{\sinh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)}.} \quad (17)$$

3.2 Second moment $\bar{M}_G^{(2)}$ (diffusion)

The second moment tensor is:

$$M_{G,ij}^{(2)} = \sum_{\bar{a}} a_i a_j \pi_G(\bar{a}). \quad (18)$$

Because every action is along a single cardinal axis, $a_x a_y = 0$ for every action, so the off-diagonal component vanishes identically: $M_{G,xy}^{(2)} = 0$.

For the diagonal xx -component, only the $\pm \hat{x}$ actions contribute (since $(\pm \hat{y})_x = 0$):

$$\bar{M}_{G,xx}^{(2)} = \epsilon^2 [\pi_G(+\hat{x}) + \pi_G(-\hat{x})] = \epsilon^2 \frac{e^{\beta u_x} + e^{-\beta u_x}}{Z_G} = \epsilon^2 \frac{2 \cosh(\beta u_x)}{2 \cosh(\beta u_x) + 2 \cosh(\beta u_y)}. \quad (19)$$

Therefore:

$$\boxed{\bar{M}_{G,xx}^{(2)} = \epsilon^2 \frac{\cosh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)}, \quad \bar{M}_{G,yy}^{(2)} = \epsilon^2 \frac{\cosh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)}}. \quad (20)$$

Consistency check:

$$\bar{M}_{G,xx}^{(2)} + \bar{M}_{G,yy}^{(2)} = \epsilon^2 \frac{\cosh(\beta u_x) + \cosh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)} = \epsilon^2. \quad \checkmark \quad (21)$$

The total second moment is always ϵ^2 regardless of goal direction; only the *anisotropy* between the two axes varies.

Relation to drift:

$$\bar{M}_{G,x}^{(1)} = \bar{M}_{G,xx}^{(2)} \tanh(\beta u_x), \quad (22)$$

which is the discrete analogue of the Einstein relation.

4 Egocentric Moments

4.1 First moment $\bar{M}_{E,x}^{(1)}$ (drift)

4.1.1 Exact discrete expression

$$\bar{M}_{E,x}^{(1)} = \epsilon [\pi_E(+\hat{x}) - \pi_E(-\hat{x})] = \epsilon \frac{e^{-\alpha O_{+x}} - e^{-\alpha O_{-x}}}{Z_E}. \quad (23)$$

Unlike the geocentric case, O_{+x} and O_{-x} are **independent** binary variables: each cell is blocked or free independently. Enumerating all four cases:

O_{+x}	O_{-x}	Numerator	Physical meaning
0	0	$1 - 1 = 0$	Both free; no bias
0	1	$1 - e^{-\alpha} > 0$	Pushed toward $+x$
1	0	$e^{-\alpha} - 1 < 0$	Pushed toward $-x$
1	1	$e^{-\alpha} - e^{-\alpha} = 0$	Both blocked; no bias

4.1.2 Continuum limit

We replace the discrete occupancy values with a smooth obstacle density field $O(\bar{r}) \in [0, 1]$, and expand the occupancies of neighbouring sites via Taylor expansion in ϵ :

$$O_{\pm x} \equiv O(\bar{r} \pm \epsilon \hat{x}) = O \pm \epsilon \partial_x O + \frac{\epsilon^2}{2} \partial_{xx} O + \mathcal{O}(\epsilon^3), \quad (24)$$

where $O \equiv O(\bar{r})$ for brevity. Expanding the exponentials:

$$\begin{aligned} e^{-\alpha O_{\pm x}} &= e^{-\alpha O} e^{\mp \alpha \epsilon \partial_x O - \frac{\alpha \epsilon^2}{2} \partial_{xx} O + \mathcal{O}(\epsilon^3)} \\ &= e^{-\alpha O} e^{-\frac{\alpha \epsilon^2}{2} \partial_{xx} O} e^{\mp \alpha \epsilon \partial_x O} + \mathcal{O}(\epsilon^3) \\ &\approx e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O\right) \left(1 \mp \alpha \epsilon \partial_x O + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2\right) + \mathcal{O}(\epsilon^3). \end{aligned} \quad (25)$$

Numerator of (23):

$$\begin{aligned}
e^{-\alpha O_{+x}} - e^{-\alpha O_{-x}} &= e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) (e^{-\alpha \epsilon \partial_x O} - e^{+\alpha \epsilon \partial_x O}) + \mathcal{O}(\epsilon^3) \\
&= e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) (-2) \sinh(\alpha \epsilon \partial_x O) + \mathcal{O}(\epsilon^3) \\
&\approx -2\alpha \epsilon e^{-\alpha O} \partial_x O + \mathcal{O}(\epsilon^3),
\end{aligned} \tag{26}$$

where in the last line we used $\sinh(z) \approx z$ for small $z = \alpha \epsilon \partial_x O$.

Denominator $Z_E = Z_{E_x} + Z_{E_y}$. First compute Z_{E_x} :

$$\begin{aligned}
Z_{E_x} &= e^{-\alpha O_{+x}} + e^{-\alpha O_{-x}} \\
&= e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) (e^{-\alpha \epsilon \partial_x O} + e^{+\alpha \epsilon \partial_x O}) + \mathcal{O}(\epsilon^3) \\
&= 2e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) \cosh(\alpha \epsilon \partial_x O) + \mathcal{O}(\epsilon^3) \\
&\approx 2e^{-\alpha O} \left(1 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) \left(1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 \right) + \mathcal{O}(\epsilon^4) \\
&\approx 2e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) + \mathcal{O}(\epsilon^4).
\end{aligned} \tag{27}$$

By the identical calculation with $x \leftrightarrow y$:

$$Z_{E_y} \approx 2e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_y O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{yy} O \right) + \mathcal{O}(\epsilon^4). \tag{28}$$

Adding:

$$Z_E = Z_{E_x} + Z_{E_y} \approx 4e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 - \frac{\alpha \epsilon^2}{4} \nabla^2 O \right) + \mathcal{O}(\epsilon^4). \tag{29}$$

Combining: The common factor $e^{-\alpha O}$ cancels in the ratio. Using $(1+z)^{-1} \approx 1-z$ for the $\mathcal{O}(\epsilon^2)$ denominator correction:

$$\bar{M}_{E,x}^{(1)} = \epsilon \frac{-2\alpha \epsilon e^{-\alpha O} \partial_x O}{4e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 - \frac{\alpha \epsilon^2}{4} \nabla^2 O \right)} = -\frac{\alpha \epsilon^2}{2} \partial_x O \left(1 - \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 + \frac{\alpha \epsilon^2}{4} \nabla^2 O \right) + \mathcal{O}(\epsilon^5). \tag{30}$$

Dropping $\mathcal{O}(\epsilon^4)$ terms:

$$\boxed{\bar{M}_{E,x}^{(1)} = -\frac{\alpha \epsilon^2}{2} \partial_x O(\bar{r}) + \mathcal{O}(\epsilon^4), \quad \bar{M}_E^{(1)} = -\frac{\alpha \epsilon^2}{2} \nabla O(\bar{r}).} \tag{31}$$

The egocentric drift is proportional to the *negative* obstacle gradient: the agent is pushed away from regions of high obstacle density.

4.2 Second moment $\bar{M}_E^{(2)}$ (diffusion)

4.2.1 Exact discrete expression

The off-diagonal component vanishes for the same geometric reason as before. For the xx diagonal:

$$\bar{M}_{E,xx}^{(2)} = \epsilon^2 [\pi_E(+\hat{x}) + \pi_E(-\hat{x})] = \epsilon^2 \frac{e^{-\alpha O_{+x}} + e^{-\alpha O_{-x}}}{Z_E} = \epsilon^2 \frac{Z_{E_x}}{Z_{E_x} + Z_{E_y}}. \tag{32}$$

The numerator is exactly Z_{E_x} — so $\bar{M}_{E,xx}^{(2)}$ is simply ϵ^2 times the *fraction of total policy weight on the x -axis actions*. Enumerating the cases:

O_{+x}	O_{-x}	Z_{E_x}
0	0	2
0	1	$1 + e^{-\alpha}$
1	0	$e^{-\alpha} + 1$
1	1	$2e^{-\alpha}$

4.2.2 Continuum limit

Using the expanded Z_{E_x} , Z_{E_y} , and Z_E derived above:

$$\begin{aligned}\bar{M}_{E,xx}^{(2)} &= \epsilon^2 \frac{Z_{E_x}}{Z_E} \\ &= \epsilon^2 \cdot \frac{2e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right)}{4e^{-\alpha O} \left(1 + \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 - \frac{\alpha \epsilon^2}{4} \nabla^2 O \right)}.\end{aligned}\quad (33)$$

The $e^{-\alpha O}$ factors cancel:

$$\bar{M}_{E,xx}^{(2)} = \frac{\epsilon^2}{2} \cdot \frac{1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O}{1 + \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 - \frac{\alpha \epsilon^2}{4} \nabla^2 O}.\quad (34)$$

Expanding the denominator using $(1+z)^{-1} \approx 1-z$ and multiplying out, keeping terms to $\mathcal{O}(\epsilon^2)$ (i.e. $\mathcal{O}(\epsilon^4)$ relative to the leading ϵ^2):

$$\begin{aligned}\bar{M}_{E,xx}^{(2)} &= \frac{\epsilon^2}{2} \left(1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O \right) \left(1 - \frac{\alpha^2 \epsilon^2}{4} |\nabla O|^2 + \frac{\alpha \epsilon^2}{4} \nabla^2 O \right) + \mathcal{O}(\epsilon^6) \\ &= \frac{\epsilon^2}{2} \left[1 + \frac{\alpha^2 \epsilon^2}{2} (\partial_x O)^2 - \frac{\alpha \epsilon^2}{2} \partial_{xx} O - \frac{\alpha^2 \epsilon^2}{4} (\partial_x O)^2 - \frac{\alpha^2 \epsilon^2}{4} (\partial_y O)^2 + \frac{\alpha \epsilon^2}{4} \nabla^2 O \right] + \mathcal{O}(\epsilon^6).\end{aligned}\quad (35)$$

Collecting terms:

$$\boxed{\bar{M}_{E,xx}^{(2)} = \frac{\epsilon^2}{2} \left[1 + \frac{\alpha^2 \epsilon^2}{4} ((\partial_x O)^2 - (\partial_y O)^2) - \frac{\alpha \epsilon^2}{4} (\partial_{xx} O - \partial_{yy} O) \right] + \mathcal{O}(\epsilon^6).}\quad (36)$$

By the identical calculation with $x \leftrightarrow y$:

$$\boxed{\bar{M}_{E,yy}^{(2)} = \frac{\epsilon^2}{2} \left[1 + \frac{\alpha^2 \epsilon^2}{4} ((\partial_y O)^2 - (\partial_x O)^2) - \frac{\alpha \epsilon^2}{4} (\partial_{yy} O - \partial_{xx} O) \right] + \mathcal{O}(\epsilon^6).}\quad (37)$$

Consistency check:

$$\bar{M}_{E,xx}^{(2)} + \bar{M}_{E,yy}^{(2)} = \epsilon^2 + \mathcal{O}(\epsilon^6). \quad \checkmark \quad (38)$$

The $\mathcal{O}(\epsilon^4)$ corrections are antisymmetric between x and y and cancel exactly. To leading order in ϵ , both components equal $\epsilon^2/2$, so the egocentric diffusion is **isotropic at leading order**; anisotropy enters only at $\mathcal{O}(\epsilon^4)$.

5 The Nonlinear Fokker-Planck Equation

5.1 Combined moments

The total drift and diffusion moments are:

$$\bar{M}_x^{(1)} = \gamma \bar{M}_{G,x}^{(1)} + (1 - \gamma) \bar{M}_{E,x}^{(1)}, \quad (39)$$

$$\bar{M}_y^{(1)} = \gamma \bar{M}_{G,y}^{(1)} + (1 - \gamma) \bar{M}_{E,y}^{(1)}, \quad (40)$$

$$\bar{M}_{xx}^{(2)} = \gamma \bar{M}_{G,xx}^{(2)} + (1 - \gamma) \bar{M}_{E,xx}^{(2)}, \quad (41)$$

$$\bar{M}_{yy}^{(2)} = \gamma \bar{M}_{G,yy}^{(2)} + (1 - \gamma) \bar{M}_{E,yy}^{(2)}. \quad (42)$$

Substituting all results, and dropping $\mathcal{O}(\epsilon^4)$ terms throughout:

Drift:

$$\bar{M}_x^{(1)} = \frac{\gamma \epsilon \sinh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)} - \frac{(1 - \gamma) \alpha \epsilon^2}{2} \partial_x O, \quad (43)$$

$$\bar{M}_y^{(1)} = \frac{\gamma \epsilon \sinh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)} - \frac{(1 - \gamma) \alpha \epsilon^2}{2} \partial_y O. \quad (44)$$

Diffusion:

$$\bar{M}_{xx}^{(2)} = \frac{\gamma \epsilon^2 \cosh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)} + \frac{(1 - \gamma) \epsilon^2}{2}, \quad (45)$$

$$\bar{M}_{yy}^{(2)} = \frac{\gamma \epsilon^2 \cosh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)} + \frac{(1 - \gamma) \epsilon^2}{2}. \quad (46)$$

Note that the $\mathcal{O}(\epsilon^4)$ anisotropy corrections in $\bar{M}_{E,x}^{(2)}$ have been dropped as they are subdominant.

5.2 Full nonlinear Fokker-Planck equation

Substituting into the Fokker-Planck equation (13):

$$\boxed{\begin{aligned} \frac{\partial P}{\partial t} = & -\partial_x \left[\left(\frac{\gamma \epsilon \sinh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)} - \frac{(1 - \gamma) \alpha \epsilon^2}{2} \partial_x O \right) P \right] \\ & -\partial_y \left[\left(\frac{\gamma \epsilon \sinh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)} - \frac{(1 - \gamma) \alpha \epsilon^2}{2} \partial_y O \right) P \right] \\ & + \frac{1}{2} \partial_{xx} \left[\left(\frac{\gamma \epsilon^2 \cosh(\beta u_x)}{\cosh(\beta u_x) + \cosh(\beta u_y)} + \frac{(1 - \gamma) \epsilon^2}{2} \right) P \right] \\ & + \frac{1}{2} \partial_{yy} \left[\left(\frac{\gamma \epsilon^2 \cosh(\beta u_y)}{\cosh(\beta u_x) + \cosh(\beta u_y)} + \frac{(1 - \gamma) \epsilon^2}{2} \right) P \right]. \end{aligned}} \quad (47)$$

The spatial dependence enters through:

- $\hat{u}(\bar{r}) = (\bar{g} - \bar{r})/|\bar{g} - \bar{r}|$, which varies smoothly with position and encodes the goal direction;
- $\nabla O(\bar{r})$, which varies with the obstacle geometry.

Physical structure. The equation has four terms:

Term	Origin	Role
$\gamma \sinh(\beta u_i)/Z_G$	Geocentric drift	Advects P toward goal
$(1 - \gamma)\alpha \nabla O/2$	Egocentric drift	Repels P from obstacles
$\gamma \cosh(\beta u_i)/Z_G$	Geocentric diffusion	Anisotropic, aligned to goal
$(1 - \gamma)/2$	Egocentric diffusion	Isotropic baseline

6 Linearized Fokker-Planck Equation

6.1 Setup: small angle approximation

Let θ be the angle of the goal direction $\hat{u}$ measured from the $+\hat{x}$ axis. We assume **small angle**: $\theta \ll 1$, meaning the agent is nearly aligned with the goal. Then:

$$u_x = \cos \theta \approx 1 - \frac{\theta^2}{2}, \quad u_y = \sin \theta \approx \theta. \quad (48)$$

6.2 Linearizing the geocentric partition function

Expand each cosh term to $\mathcal{O}(\theta^2)$:

$$\cosh(\beta u_x) = \cosh\left(\beta - \frac{\beta\theta^2}{2}\right) \approx \cosh \beta - \frac{\beta\theta^2}{2} \sinh \beta, \quad (49)$$

$$\cosh(\beta u_y) = \cosh(\beta\theta) \approx 1 + \frac{\beta^2\theta^2}{2}. \quad (50)$$

Therefore:

$$Z_G \approx 2(1 + \cosh \beta) + \beta\theta^2(\beta - \sinh \beta) = 2(1 + \cosh \beta) \left[1 + \frac{\beta\theta^2(\beta - \sinh \beta)}{2(1 + \cosh \beta)} \right]. \quad (51)$$

6.3 Linearized geocentric drift

x -component. Expand $\sinh(\beta u_x)$:

$$\sinh(\beta u_x) = \sinh\left(\beta - \frac{\beta\theta^2}{2}\right) \approx \sinh \beta - \frac{\beta\theta^2}{2} \cosh \beta. \quad (52)$$

Substituting and expanding $(1 + z)^{-1} \approx 1 - z$ for the denominator:

$$\bar{M}_{G,x}^{(1)} \approx \frac{\epsilon \sinh \beta}{2(1 + \cosh \beta)} \left[1 - \frac{\beta\theta^2}{2} \left(\coth \beta + \frac{\beta - \sinh \beta}{1 + \cosh \beta} \right) \right]. \quad (53)$$

Using the identity $\sinh \beta / (1 + \cosh \beta) = \tanh(\beta/2)$, the leading term is $\epsilon \tanh(\beta/2)$.

y-component. Expand $\sinh(\beta u_y) = \sinh(\beta \theta) \approx \beta \theta$:

$$\bar{M}_{G,y}^{(1)} \approx \frac{\epsilon \beta \theta}{2(1 + \cosh \beta)} + \mathcal{O}(\theta^3). \quad (54)$$

Using $1 + \cosh \beta = 2 \cosh^2(\beta/2)$:

$$\bar{M}_{G,y}^{(1)} \approx \frac{\epsilon \beta \theta}{4 \cosh^2(\beta/2)}. \quad (55)$$

6.4 Linearized geocentric diffusion

xx-component.

$$\bar{M}_{G,xx}^{(2)} \approx \frac{\epsilon^2 \cosh \beta}{2(1 + \cosh \beta)} \left[1 - \frac{\beta \theta^2}{2} \left(\tanh \beta + \frac{\beta - \sinh \beta}{1 + \cosh \beta} \right) \right]. \quad (56)$$

yy-component. Using $\cosh(\beta u_y) \approx 1 + \beta^2 \theta^2 / 2$:

$$\bar{M}_{G,yy}^{(2)} \approx \frac{\epsilon^2}{2(1 + \cosh \beta)} \left[1 + \frac{\beta^2 \theta^2}{2} - \frac{\beta \theta^2 (\beta - \sinh \beta)}{2(1 + \cosh \beta)} \right]. \quad (57)$$

The $\mathcal{O}(\theta^2)$ corrections to the diffusion tensor multiply second derivatives of P and therefore contribute at $\mathcal{O}(\theta^2)$ relative to the leading diffusion terms. We drop these higher-order corrections and retain only the leading-order, θ -independent diffusion.

6.5 Egocentric moments at leading order

The egocentric drift is already linear in the obstacle gradient and requires no further expansion in θ . Dropping $\mathcal{O}(\epsilon^4)$:

$$\bar{M}_{E,x}^{(1)} = -\frac{\alpha \epsilon^2}{2} \partial_x O, \quad \bar{M}_{E,y}^{(1)} = -\frac{\alpha \epsilon^2}{2} \partial_y O, \quad (58)$$

$$\bar{M}_{E,xx}^{(2)} = \frac{\epsilon^2}{2}, \quad \bar{M}_{E,yy}^{(2)} = \frac{\epsilon^2}{2}. \quad (59)$$

6.6 Defining velocity and diffusion constants

Collecting all terms at leading order in θ and ϵ , define the following position-independent constants:

$$v_0 = \gamma\epsilon \tanh \frac{\beta}{2} = \frac{\gamma\epsilon \sinh \beta}{2(1 + \cosh \beta)}, \quad (60)$$

$$\kappa = \frac{\gamma\epsilon\beta}{4 \cosh^2(\beta/2)} = \frac{\gamma\epsilon\beta}{2(1 + \cosh \beta)}, \quad (61)$$

$$D_{\parallel} = \frac{\gamma\epsilon^2 \cosh \beta}{2(1 + \cosh \beta)} + \frac{(1 - \gamma)\epsilon^2}{2}, \quad (62)$$

$$D_{\perp} = \frac{\gamma\epsilon^2}{2(1 + \cosh \beta)} + \frac{(1 - \gamma)\epsilon^2}{2}. \quad (63)$$

These have clear physical meanings:

Symbol	Expression	Meaning
v_0	$\gamma\epsilon \tanh(\beta/2)$	Forward speed along goal axis
κ	$\gamma\epsilon\beta/4 \cosh^2(\beta/2)$	Transverse restoring strength
$D_{\parallel}$	Eq. (62)	Diffusion along goal axis
$D_{\perp}$	Eq. (63)	Diffusion transverse to goal

The combined linearised velocity field is:

$$V_x(\bar{r}) = v_0 - \frac{(1 - \gamma)\alpha\epsilon^2}{2} \partial_x O(\bar{r}), \quad (64)$$

$$V_y(\bar{r}) = \kappa \theta(\bar{r}) - \frac{(1 - \gamma)\alpha\epsilon^2}{2} \partial_y O(\bar{r}). \quad (65)$$

Note that $\theta(\bar{r}) = \arctan((g_y - y)/(g_x - x)) \approx (g_y - y)/(g_x - x)$ for small angles, so V_y contains a linear restoring force toward the goal axis.

6.7 Linearized Fokker-Planck equation

Substituting the linearised velocity and diffusion constants into the Fokker-Planck equation, and dropping all $\mathcal{O}(\epsilon^4)$ and $\mathcal{O}(\theta^2)$ corrections:

$$\boxed{\frac{\partial P}{\partial t} = -\partial_x [V_x(\bar{r}) P] - \partial_y [V_y(\bar{r}) P] + D_{\parallel} \partial_{xx} P + D_{\perp} \partial_{yy} P,} \quad (66)$$

with V_x, V_y given by (64)–(65) and diffusion constants by (62)–(63).

6.8 Physical interpretation

Each term in the linearized FP equation has a distinct physical role:

Term	Expression	Role
Forward advection	$-v_0 \partial_x P$	Sweeps P along goal axis at speed $v_0 = \gamma \epsilon \tanh(\beta/2)$
Transverse restoring	$-\kappa \partial_y(\theta P)$	Linear-in- θ force; realigns P toward goal direction
Obstacle repulsion	$+\frac{(1-\gamma)\alpha\epsilon^2}{2} \nabla \cdot (\nabla O \cdot P)$	Pushes P away from high obstacle density
Forward diffusion	$D_{\parallel} \partial_{xx} P$	Spreads P along goal axis
Transverse diffusion	$D_{\perp} \partial_{yy} P$	Spreads P perpendicular to goal

6.9 Limiting behavior of the diffusion constants

Isotropic limit ($\beta \rightarrow 0$):

$$D_{\parallel} \rightarrow \frac{\epsilon^2(2-\gamma)}{4} = D_{\perp}, \quad (67)$$

so diffusion is isotropic and both axes have equal spreading.

Strongly anisotropic limit ($\beta \rightarrow \infty$):

$$D_{\parallel} \rightarrow \frac{\epsilon^2}{2}, \quad D_{\perp} \rightarrow \frac{(1-\gamma)\epsilon^2}{2}, \quad (68)$$

so the anisotropy ratio is:

$$\left. \frac{D_{\parallel}}{D_{\perp}} \right|_{\gamma=1} = \cosh \beta, \quad (69)$$

which grows without bound with β : the agent diffuses strongly along the goal axis and barely at all in the transverse direction.

Pure egocentric ($\gamma = 0$):

$$D_{\parallel} = D_{\perp} = \frac{\epsilon^2}{2}, \quad (70)$$

giving isotropic diffusion with no goal-induced anisotropy.